Parley

INDUSTRIAL AGENTS HACKATHON · MUNICH 2026 · TRACK TACTO

DEAL LOCKED

PARLEY.

THE AI BUYER THAT NEGOTIATES.

Code does the math, AI does the talking: verified facts and equal-cost trades, built on game theory.



TOTAL COST

Track **Tacto Negotiation Arena**	Team **Clooless**	**parley.sorevo.de**

PARLEY | SHEET 01/10 | REV D

PROBLEM

EVERY PRICE INCREASE STARTS AS AN E-MAIL.

Gießerei Brandlhuber GmbH & Co. KG

PRACTICE CASE P01

PRICE ADJUSTMENT FROM 01 NOVEMBER 2026

Owing to sustained increases in energy, raw material and wage costs, we are forced to adjust our prices for volute casing HN 50-200 by **9.5 %**: from EUR 126.53 to EUR 138.55 per piece.

to: Purchasing, Hollerau Pumpenwerke

TO ANSWER IT WELL, A BUYER CHECKS

3 years of orders, invoices and scorecards

33 market price indices

447 suppliers, each with notes, news and contracts

> The supplier checks every argument against the same data. **Only solid facts move the price.**

Source: supplier letter of 11 Sep 2026, practice case P01; Tacto Negotiation Arena data package (33,000 rows in 24 tables).

READS. NEGOTIATES. SIGNS OR WALKS.

01 · READ

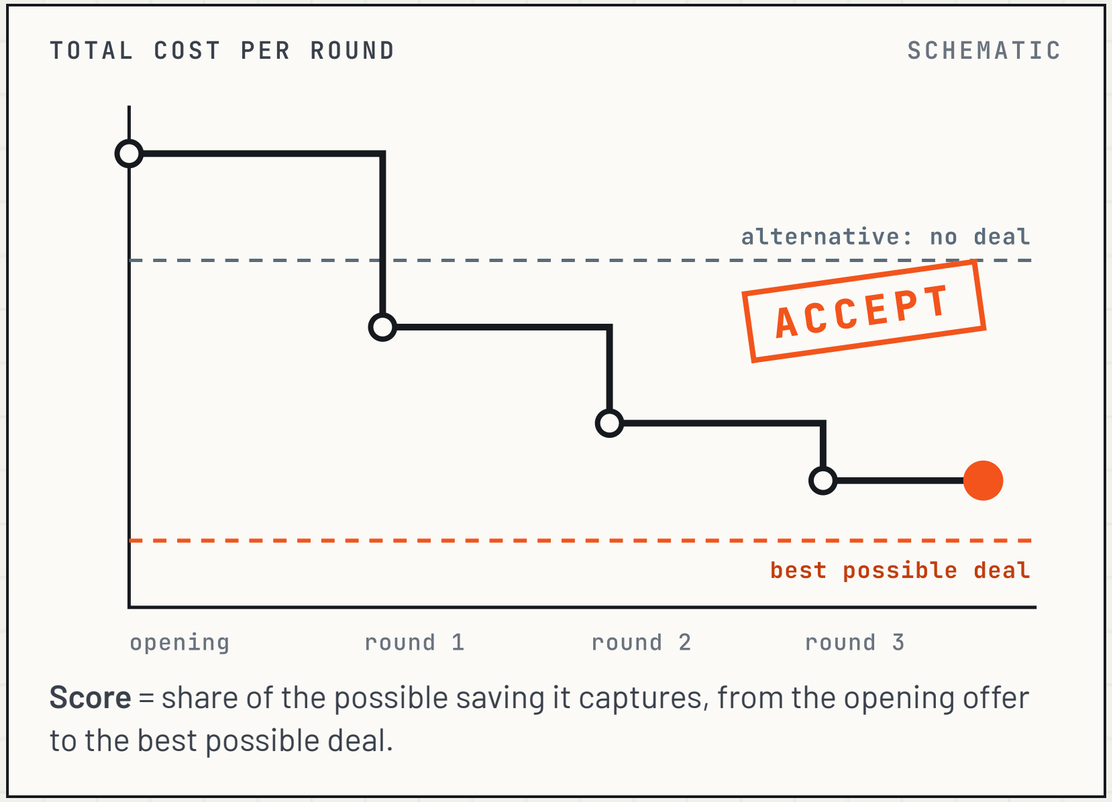
The e-mail, the quote PDF and three years of history with this supplier.

02 · NEGOTIATE

Up to ten e-mails. Every offer becomes one **total cost** over the contract term.

03 · CLOSE

Accept the cheapest offer that keeps every must-have, or walk away when the alternative is cheaper.



Score = share of the possible saving it captures, from the opening offer to the best possible deal.

HOW IT WORKS

FACTS. PRICED CHOICES. A LOGGED DECISION.

01 · CODE CHECKS

SUPPLIER FACTS

Each with its source ID. Profiles built offline, checked by code; we keep only what passes our gate.

02 · CODE PRICES

PRICED CHOICES

Two or three packages per round at about the same estimated total cost. The supplier's pick shows what it values; our steps answer its steps.

03 · CODE LOGS

LOGGED DECISION

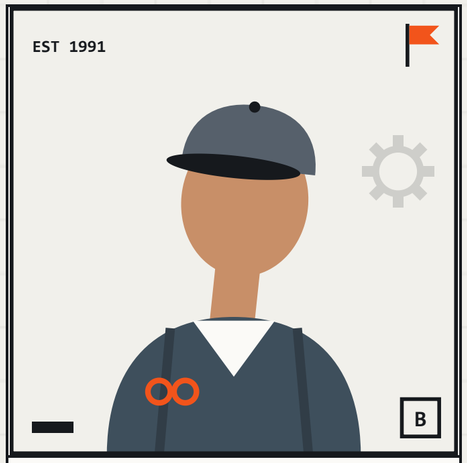
Accept, hold or walk, with the reason. Templates write the e-mail; the model stays off.

EVERY E-MAIL On-time delivery averaged `81.8 %` `sc:70210@2026Q3`. Package 1: `€28.58`, `14` days net, `35` days lead time.

☐ TEMPLATE ☐ CODE

Game-theory-inspired bargaining: preference probing and bounded reciprocal concessions (agent/policy.py, packages()). Details: backup B1.

KNOWS THE SUPPLIER. MOVES THE PRICE.

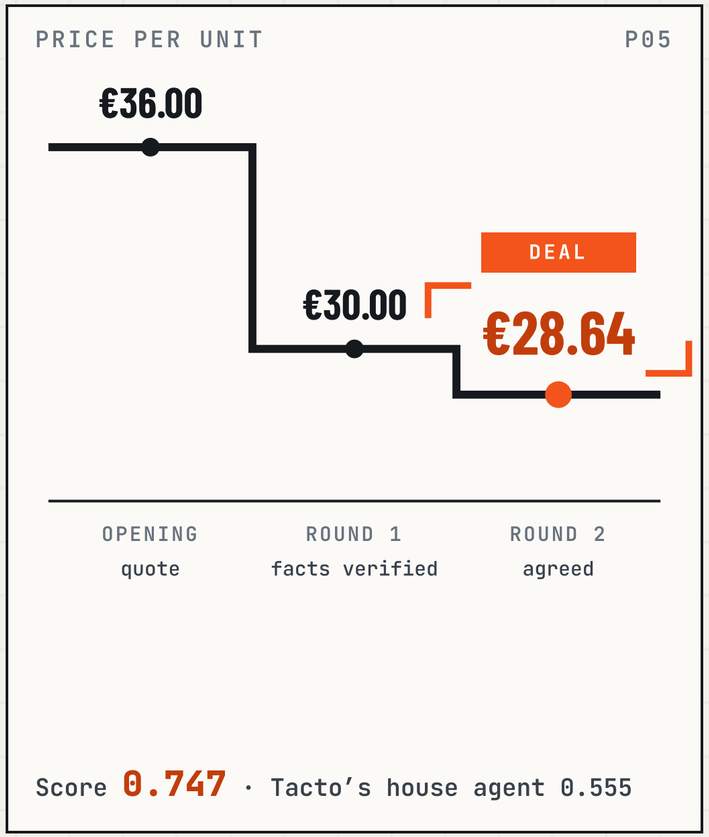


ANDREAS DRALVECK

Owner · Zerspanung Dralveck GmbH

22-person machining shop, founder-owned; the owner negotiates himself

PARLEY KNOWS	SO IT
A rival quotes €31.50 per unit `offer:70240-0`	Cites the rival quote with its ID
Has not tracked his material prices since the 2022 peak `doc:20260908-70210-M1`	Cites the index that fell: AS-14404 –8.5 %
Second shift on hold since spring `doc:20260722-70210-M2`	Offers binding volume: 4,200 pcs a year, 24 months
Complained about overdue invoices `doc:20260504-70210-P`	Offers 14- or 30-day payment, priced into total cost

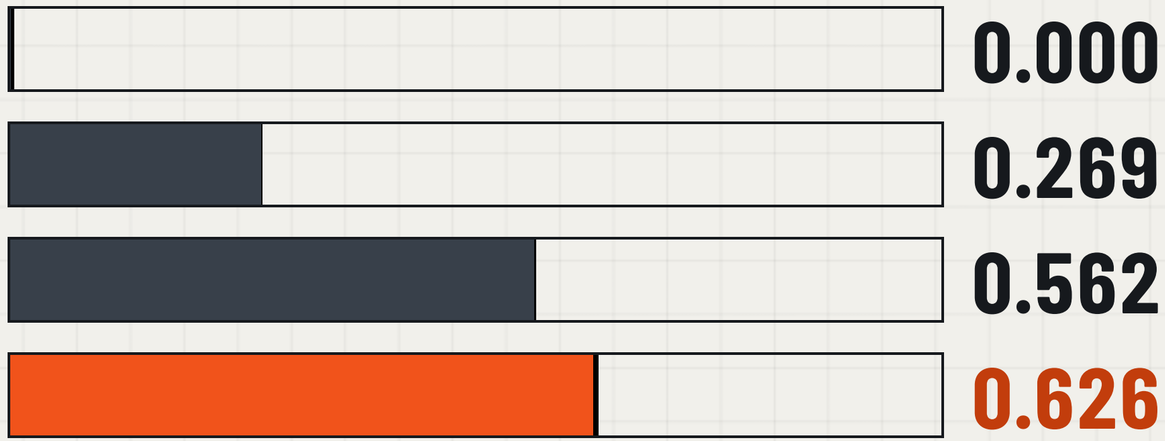


Source: our Saturday practice session P05, Parley v1, Sat 24 Oct (replay e642aeb72a327558). Round 1 offered €28.58 paid in 14 days or €28.68 in 30 days, both 35-day delivery, similar estimated cost. Figures and packages from code; the final build writes every e-mail from templates. Dossier: Tacto's data package. Supplier and portrait fictional.

FROM OUR OWN PRACTICE RUNS · CASE P02

OUR ZERO. FOUND, FIXED, TESTED.

PRACTICE CASE P02 · ARENA SCORE BY BUILD



Strategic supplier: the brief says a healthy supplier counts, not only the lowest price.

WHAT WENT WRONG

Price it accepted	€340.94
Supplier-health charge	€224,039
Total cost vs. opening	higher · score 0
Our cost vs. arena, now	✓ within 1 %

The fix: price the supplier's full cost, not just the unit price. A test fails if our cost drifts from the arena's.

Source: our arena practice runs on P02 (our arena practice log): v1.16 029552b2c280199e, v1.22 0885001f42db6d2e, v1.30 da71ec8f9b9db3e1, final build settings ebe0c9d87c1f3cc8; arena cost breakdown of the v1.16 deal; agent test test_our_cost_matches_the_arena_including_the_supplier_health_charge.

WHY DID IT WRITE THAT? IT SAYS.

PARLEY → DRALVECK — ROUND 1

```
anchor at 28.80 on the standing structure (20.0% below)
```

Cites `offer:70240-0` `idx:AS-14404` `sc:70210`. Three packages, about the same estimated cost for us: **€28.58**, €28.68, €28.41.

PARLEY → DRALVECK — ROUND 2

```
move 4% of the gap (the supplier announced an internal step: hold)
```

He said he had to check internally. Two packages: **€28.64**, €28.74.

PARLEY · CLOSE — ROUND 3

```
every thread has finished; accept the cheapest standing offer
C=244,164
```

Accepts **€28.64**, cheapest by total cost over 24 months.

AUDIT TRAIL

Facts cited, data IDs	✓ 3 verified
Packages, ≈ same est. cost	3 → 2
Engine	deterministic
Same inputs	same decision

On /replay: the replay shows the reason for each buyer move it recorded, with the packages and the data IDs it cited.

Source: decision journal of our Saturday practice session P05, Parley v1, Sat 24 Oct (replay e642aeb72a327558); reasons verbatim, e-mails condensed. Supplier fictional.

RESULTS · PRACTICE, SAT 24 – SUN 25 OCT

MEASURED IN PRACTICE.

0.792

equal-weight mean over ten practice cases, v1.38 candidate, no model, one session each, no zeros

0.850

on seven cases

0.658

on the three earlier-labelled hold-outs

MODEL TEST · V1.9	90 paired games, full model minus none +0.0001	worst-5 % decision time, without / with 0.21 s / 7.43 s	final: buyer model calls 0

Identical repeats show it reproduces, not that it is independently validated. Source: our arena practice runs (agent/MERGES.md, v1.38 live record); ablation from the v1.9 test, not measured on v1.38 (README.md).

REAL SUPPLIERS

BUILT FOR SUPPLIERS IT HAS NEVER MET.

- **No memorised bots.** No rule names a supplier; a test fails if one does.
- **Reads the supplier live:** step sizes, final-offer timing, signal sentences.
- **Safe by construction:** accepts only below the cheapest opening, every must-have kept, cheaper than the alternative.
- **Tested on the unknown:** held-out generated cases, suppliers voiced by Claude, Gemini, GPT and DeepSeek.
- **Reads new data:** Sunday's release (supplier news and letters) became dossier facts: cash need, free capacity, cited by ID.

RULES IT PLAYS BY

- ✓ **ROUND ONE IS THE NEGOTIATION.**
- ✓ **LEAD WITH THE COST DRIVERS THAT FELL.**
- ✓ **NEVER PUSH AFTER A FINAL OFFER.**
- ✓ **WALK AWAY WHEN THE ALTERNATIVE IS CHEAPER.**

PARLEY

LOWEST TOTAL COST. A REASON FOR EVERY TRADE.

BUYER SUPPLIER

DEAL LOCKED

01 **Watch it:** every decision is logged with its reason; our runs are in the replays.

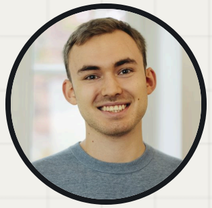
02 **Try it:** demo, technical video and architecture at parley.sorevo.de.

03 **Team Clooless:** built in 25 hours in Munich.

SV Saket Veerubhotla

TT Tanay Tiwari

TN Tanmay Narang

 Niklas Wichter

PARLEY.SOREVO.DE

Demo, videos and replays

CODE OWNS EVERY NUMBER. THE NUMBERS WIN.

CODE

DOSSIER

Supplier profile, cost floor, and facts with checked data IDs.

→

CODE

VALUATION

Prices every combination of terms as total cost, and finds sheets at about the same estimated cost.

→

CODE

POLICY

Anchor, concession steps, guards G1–G9, accept or walk away.

→

CODE

READER

Reads the reply: status, verdicts, signal sentences.

→

MODEL · OFF

WRITER

No measurable gain in our paired tests, so the final runs on templates.

WHAT WE OPTIMISE

ARENA SCORE $= \dfrac{C_{open} - C_{deal}}{C_{open} - C_{best}}$

Share of the possible saving. **C** = total cost over the contract horizon. **Hard:** every must-have kept, never worse than the opening, walk if the alternative is cheaper. **Tie:** fewer rounds, in 8-minute waves.

C COUNTS	price × volume	uncovered need + risk	payment interest	freight (Incoterm)	late lead time	defects – chargebacks	tooling	index path + risk	supplier margin

LEVERS	price	payment days	lead time	contract months	volume	effective date	index clause	ppm · chargeback

BACKUP · FROM OUR OWN PRACTICE RUN · CASE P11

FACTS CHECKED. THEN IT MOVED.

SUPPLIER · ODLEWNIA KALWISZ — ROUND 0

We are happy to present our offer: **€47.00** per unit, quality commitment **6,400 ppm**.

PARLEY · HOLLERAU PURCHASING — ROUND 1

Your on-time delivery averaged **70.8 %** over the last two quarters `sc:70140@2026Q3`, plus two price indices that fell. Three packages, each equal cost for us.

SUPPLIER · ODLEWNIA KALWISZ — ROUND 1

We checked the delivery record data you cited; **it holds.** Our terms: €46.50 per unit, **300 ppm**.

THE DECISION JOURNAL

3 claims, data IDs	✓ verified
ppm_commitment	~~6,400~~ 300
unit_price, rounds 0-3	~~47.00~~ 45.50

PACKAGE 1	PACKAGE 2
€43.13	**€43.17**
300 ppm, no chargeback	300 ppm, €150 chargeback
C €712,400	C €712,320

Logged reason:

anchor at 42.30 on the standing structure (10.0% below)

Source: Parley v1 practice session, Sat 24 Oct, decision journal and arena replay. E-mails condensed.